

2.3 Of DRM And Communication

While graphics are grouped under the Windows Graphics Foundation, Vista also groups all communications services under a single banner called the Windows Communication Foundation (WCF). Communication here includes all application-to-application messaging such as asynchronous messaging between applications or distributed transactions and Web services. These features were present in previous versions, but Vista combines them under a single banner and makes working with them easier.

Microsoft has also exposed some workflow features under the heading of Windows Workflow Foundation (WF)—these are enterprise-class tasks. (A workflow is a task that requires decisions to be taken at various points and has a flow—say a branching task to process inventory.) WF not only exposes tools and APIs to easily set up such workflows, but also tracks and monitors the workflow and is involved in task scheduling. While the WGF is meant for Vista only, WCF and WF will be back ported to Windows XP to ease developers into this new mode of working with and creating applications. Finally, the trio of WGF, WCF and WF all combine under a single parent umbrella termed the WinFX, which is the brand name for the entire .NET framework. If you are a Windows developer, your future thus lies in learning and mastering .NET tools and getting familiar with the intricacies of managed code.

Vista also makes some low-level changes to the way multimedia is handled and presented by the operating system. The biggest change panders to the movie and the music industry by introducing elements for Digital Rights Management (DRM). Typically, these involve reducing the quality of a video or audio source if your hardware does not play nice with DRM requirements. These DRM restrictions are introduced and enforced by a media layer called the Media Foundation. In previous Windows versions, DirectShow was called for to support multimedia